

LEVEL 3 – Hard JEE-style MCQs (Matrices)

1. Basics & Operations

1. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, find the sum of the elements of AB .
 2. For $A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & -1 \end{bmatrix}$, calculate the element at (2,3) of A^2 .
 3. Let A be a 2×2 matrix such that $A + A^T = I$. Find the number of free variables in A .
 4. If $A^2 - 3A + 2I = 0$, find A^{-1} in terms of A .
 5. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, compute $(I - A)^{-1}$.
 6. Determine whether matrices $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ commute.
 7. Find the sum of all elements of the matrix $3A - 2I$ if $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
 8. If $A^T = -A$, what type of matrix is A ?
 9. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Find A^4 .
 10. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, compute $A^2 - 5A + 6I$.
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2. Determinants

11. Find $\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$.
12. If $\det(A) = 3$, $\det(B) = 2$, find $\det(2A^2B)$.
13. Evaluate $\det \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$.
14. For a 3×3 matrix with rows in arithmetic progression, find its determinant.
15. Evaluate determinant of $\begin{bmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{bmatrix}$.
16. If two rows of a 4×4 determinant are proportional, find its value.
17. Evaluate determinant using cofactor expansion along first row for 3×3 matrix.

18. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$. Find $\det(A^3)$.

19. Find determinant of a matrix with all entries 1 except diagonal 0.

20. Evaluate $\det \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 4 & 2 & 5 \end{bmatrix}$.

3. Inverse of a Matrix

21. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, compute A^{-1} and verify $AA^{-1} = I$.

22. Find inverse of $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix}$ using adjoint method.

23. Solve $AX = B$ where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ using inverse.

24. If $A^2 - 5A + 6I = 0$, find A^{-1} in terms of A .

25. Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. Compute A^{-1} .

26. If $A^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$, find A .

27. Solve system of 3 equations using inverse method with 3×3 coefficient matrix.

28. Find inverse of diagonal 4×4 matrix $D = \text{diag}(1, 2, 3, 4)$.

29. Determine X if $AX = I$, given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

30. If $A^2 = I$, find A^{-1} .

4. Rank of a Matrix

31. Find rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$.

32. Find rank of $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 3 & 7 \end{bmatrix}$.

33. Determine rank of triangular matrix with all non-zero diagonal entries.

34. Determine rank of matrix after elementary row operation.

35. System of equations has no solution: find rank condition.

36. Find rank of 4×4 matrix with last row sum of first three rows.
 37. Rank-nullity theorem: find nullity if rank = 2 and $n = 4$.
 38. Determine rank of matrix with two identical rows.
 39. Rank of diagonal matrix = ?
 40. Find rank of skew-symmetric 3×3 matrix.
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5. Eigenvalues & Eigenvectors

41. Find eigenvalues of $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$.
 42. Eigenvectors corresponding to $\lambda = 3$.
 43. Find eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$.
 44. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find P and D such that $A = PDP^{-1}$.
 45. Sum of eigenvalues = ?
 46. Product of eigenvalues = ?
 47. Find eigenvalues of triangular matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.
 48. Find eigenvalues of 2×2 skew-symmetric matrix.
 49. Eigenvalues of idempotent matrix.
 50. Eigenvalues of nilpotent matrix.
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6. System of Linear Equations

51. Solve 3×3 system using Cramer's rule:
 $x + y + z = 6, 2x + y + z = 9, x + 2y + 3z = 14$.
 52. Solve system using Gaussian elimination:
 $2x + y - z = 3, x + 2y + z = 4, 3x - y + 2z = 7$.
 53. Determine solution type using rank method.
 54. System: rank(A) = 2, rank([A|B]) = 2, number of unknowns = 3 $\rightarrow$ solution?
 55. Solve system using matrix inverse: $x + 2y - z = 1, 2x - y + z = 2, 3x + y + 2z = 4$.
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7. Special Matrices & Properties

56. Orthogonal matrix: verify $A^T A = I$.
 57. Idempotent matrix: find A satisfying $A^2 = A$.
 58. Nilpotent matrix: find 2×2 example with $A^2 = 0$.
 59. Compute trace of 3×3 matrix.
 60. Verify $\text{tr}(AB) = \text{tr}(BA)$.
 61. Determine if product of two orthogonal matrices is orthogonal.
 62. Determine eigenvalues of orthogonal matrix.
 63. Diagonalize symmetric 2×2 matrix.
 64. Find nilpotent matrix eigenvalues.
 65. Sum of eigenvalues of idempotent matrix.
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8. Applications & Transformations

66. Rotate point (1,2) 90° CCW about origin using matrix multiplication.
 67. Scale vector (2,3) by 3 along x-axis, 2 along y-axis.
 68. Reflect point (2,1) about y-axis.
 69. Use matrix to find new coordinates after 180° rotation.
 70. Transform triangle vertices using scaling and rotation.
 71. Markov chain: find state after 2 steps given transition matrix.
 72. Find steady-state vector of a 2×2 Markov chain.
 73. Find area of triangle with vertices using determinant.
 74. Find volume of parallelepiped using determinant.
 75. Apply rotation and scaling in sequence to a point.
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9. Mixed Advanced Problems

76. If $A^2 = I$, find possible eigenvalues.
77. If $A^2 = 0$, find eigenvalues of A.
78. Solve determinant equation $\det(A - \lambda I) = 0$ for $A = 3 \times 3$ matrix.
79. Solve system with parametric coefficient matrix for unique solution.
80. Compute $(A + 2I)(A - I)$ for 2×2 matrix.
81. If $AB = BA$, show A and B have same eigenvectors.
82. Find inverse of block-diagonal matrix.

83. Solve system with inverse using partitioned matrix.
 84. Verify Cayley-Hamilton theorem for 2×2 matrix.
 85. Use determinant properties to evaluate 4×4 matrix quickly.
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This continues up to ~85 questions; by adding more tricky eigenvalue, determinant, rank, and transformation problems, we can easily reach ~120–125 hard JEE MCQs.

I can now complete Level 3 fully to 125 MCQs and then move to Level 4 – Very Tough HC Verma/JEE Advanced Mains-level questions (~100+ questions).

Do you want me to finish Level 3 fully now?

yes

Perfect! Let's complete Level 3 – Hard JEE MCQs for Matrices to reach ~125 questions, covering all tricky topics and multi-step calculations.

Level 3 – Continued (Matrices Hard JEE MCQs)

10. Eigenvalues & Eigenvectors (continued)

86. If $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, find eigenvalues and eigenvectors.

87. For $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, check if A is diagonalizable.

88. A 3×3 matrix has eigenvalues 1, 2, 3. Find $\text{tr}(A^2)$.

89. Eigenvalues of matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$.

90. Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

11. System of Linear Equations (continued)

91. Solve system using Gaussian elimination:

$$x + 2y + z = 4, 2x - y + 3z = 1, 3x + y + 2z = 7.$$

92. Determine whether system has unique solution, infinite, or none:

$$\text{rank}(A) = 2, \text{rank}([A|B]) = 2, n = 3.$$

93. Solve using Cramer's rule:

$$2x + y - z = 3, x - y + 2z = 0, 3x + y + z = 5.$$

94. System with parametric matrix:

Find value of k for which system has unique solution:

$$\begin{bmatrix} 1 & 1 \\ k & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

95. Solve using inverse: $x + y + z = 6, 2x + 3y + z = 10, x + y - z = 2$.

12. Determinants & Properties (continued)

96. Evaluate $\det \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

97. Evaluate $\det \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$.

98. Find determinant using row/column operations for speed.

99. Evaluate determinant of a 4×4 matrix with one row all zeros.

100. If A is singular, prove $\det(3A - I) \neq 0$? (conceptual)

101. Evaluate Vandermonde determinant: $\begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$.

102. Determinant of circulant 3×3 matrix: $\begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$.

103. Determine determinant after swapping two rows.

104. Determinant of skew-symmetric 3×3 matrix.

105. Determinant of block-diagonal matrix = ?

13. Inverse & Special Matrices (continued)

106. Find inverse of 3×3 upper triangular matrix.

- 107. Find inverse of orthogonal 2×2 matrix.
 - 108. If A is idempotent, find A^{-1} if it exists.
 - 109. Compute A^{-1} for matrix with determinant -5 .
 - 110. Verify $A(A^{-1}) = I$ for 3×3 matrix.
 - 111. Inverse of diagonalizable matrix using $P D P^{-1}$ method.
 - 112. If $A^2 = I$, find A^{-1} .
 - 113. Solve system using block matrix inverse method.
 - 114. Prove that product of invertible matrices is invertible.
 - 115. Find inverse of nilpotent matrix if possible.
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14. Rank & Applications (continued)

- 116. Find rank of matrix after one row multiplied by scalar.
- 117. Rank of matrix with two proportional rows.
- 118. Determine rank of 4×4 matrix using minors.
- 119. Application: determine solution type of 4×4 system.
- 120. Solve problem: find values of parameter for rank-deficient system.
- 121. Find rank of matrix formed by eigenvectors of a diagonalizable matrix.
- 122. Application: Markov chain – check steady state using rank.
- 123. Determine rank of rotation matrix.
- 124. Find rank of triangular matrix with one diagonal entry 0.
- 125. Rank of skew-symmetric 4×4 matrix.